

Data Analysis & Predictive Modelling Report

Title: Global Financial Performance & Predictive Sales Analysis

Tools Used: Python (Pandas, Scikit-Learn, Matplotlib, Seaborn), Power BI Desktop

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1. Executive Summary

This report provides a comprehensive evaluation of global business operations over a multi-year period (2013–2014). Utilizing a dataset of 700 transaction records, we performed Exploratory Data Analysis (EDA) to understand seasonal patterns and geographic distributions, built a Multiple Linear Regression model to forecast sales volumes, and deployed an interactive Power BI dashboard to present key performance metrics to stakeholders.

The business demonstrated strong commercial expansion, generating **\$118.73M in Total Sales**, moving **1.13M units**, and securing **\$16.89M in Net Profit** (an overall profit margin of **14.2%**). Year-over-year revenue escalated heavily from \$26.42M (2013) to \$92.31M (2014).

2. Methodology & Implementation

Phase 1: Exploratory Data Analysis (EDA)

Data integrity was established by stripping whitespace from columns and checking for null values (isolated primarily in the Discount Band column).

- **Distribution Check:** Univariate analysis showed that both Sales and Profit metrics are heavily right-skewed, a classic footprint of a retail/b2b structure consisting of frequent low-value transactions balanced by select high-value contract closures.
- **Feature Relationships:** Bivariate and correlation metrics revealed near-perfect mathematical correlation ($r = 1.0$ apx.) between Sales, Gross Sales, and COGS.

Phase 2: Predictive Modelling (Regression)

To engineer a realistic predictive framework, variables directly containing post-transaction financial summaries (Gross Sales, COGS, Profit) were excluded to prevent data leakage and perfect multicollinearity.

The model targeted the Sales variable using operational features: **Units Sold, Sale Price, Segment, and Country**. Categorical features were converted via One-Hot Encoding (`pd.get_dummies`), and data was processed using an 80/20 train-test split via LinearRegression in Scikit-Learn.

Phase 3: Business Intelligence (Power BI Dashboard)

An interactive executive dashboard was constructed using a strategic information hierarchy:

1. **High-Level KPI Blocks:** Total Sales, Profit, and Volume explicitly pinned to the top header.
2. **Temporal Trends:** A dual-line timeline chart capturing revenue trajectory variations across months.
3. **Categorical & Regional Share:** A donut chart illustrating product performance share paired with vertical bar charts tracking sales by country.
4. **Interactive Layer:** Dynamic slicers filtering data instantly by Segment and Year.

3. Core Insights & Financial Performance

Market Segment Analysis

- **Government (The Core Driver):** Generating **\$52.50M (44.2%)** of total revenue, this segment is highly optimized, carrying a strong **21.7% net profit margin**.
- **Channel Partners (The Hidden Margin Asset):** While representing a small portion of gross sales (\$1.80M), it operates at an extraordinary **73.1% profit margin**, yielding \$1.32M in pure profit due to highly efficient, low-overhead transaction parameters.
- **Enterprise (The Financial Risk Area):** The segment brought in a substantial \$19.61M in volume but generated a net loss of **-\$614,545 (-3.1% margin)**.

Product & Regional Standouts

- **Product Performance: Paseo** is the clear flagship product line for the enterprise, driving **\$33.01M** in gross sales volume.
- **Geographic Distribution:** Revenue is exceptionally balanced across modern international boundaries, led by the **United States (\$25.03M)** and followed closely by Canada (\$24.89M), France (\$24.35M), and Germany (\$23.51M).

4. Strategic Recommendations for Stakeholders

1. **Fix the Enterprise Profit Drain:** Management must immediately audit the price structures, manufacturing costs (COGS), or aggressive discount bands applied to the Enterprise segment to stop capital erosion.
2. **Optimize U.S. Pricing Schemes:** The United States generates our highest top-line sales volumes (\$25.03M) but holds our lowest national margin (**11.9%**). We recommend adopting the pricing and distribution models used in France and Germany, which successfully yield superior capital efficiency (>15.5% margins).
3. **Scale the Channel Partner Model:** Because Channel Partners boast a 73.1% margin profile, even modest increases in marketing and distribution allocations to this channel will directly and significantly boost bottom-line net profit.